

Semiconductors & PN Junction Diode

Detailed Exam Notes with Diagrams

PHYSICS / ELECTRONICS — REVISION NOTES

Compiled with concept diagrams, comparison charts, and last-minute revision summary

1. Introduction to Semiconductors

A **semiconductor** is a material whose electrical conductivity lies between that of a good conductor and an insulator.

Semiconductor conductivity can be controlled by several factors:

- Temperature
- Addition of impurities (doping)
- Applied electric field
- Light (in some semiconductor devices)

Important property: Unlike metals, the conductivity of a semiconductor generally **increases** when temperature increases, because more charge carriers become available for conduction.

Common semiconductor materials: Silicon (Si) and Germanium (Ge). Silicon is used most widely in practical electronic devices.

Conductivity Spectrum

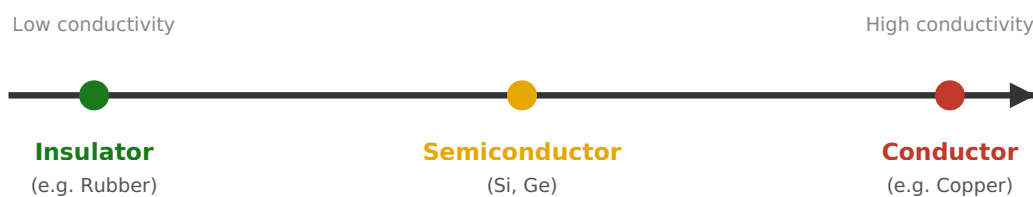


Fig 1.1 — Semiconductors occupy the conductivity range between insulators and conductors.

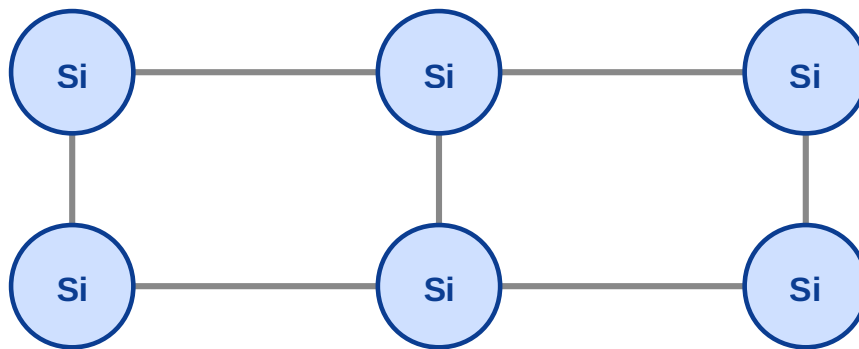
2. Intrinsic Semiconductor

An **intrinsic semiconductor** is a **PURE** semiconductor with no intentionally added impurity (e.g. pure silicon, pure germanium).

- Electrons and holes are generated in pairs.
- Number of free electrons = number of holes.
- Conductivity is relatively low compared to a doped semiconductor.

 **MEMORY TIP: INTRINSIC = PURE**

Intrinsic Silicon Crystal Lattice



m shares 4 valence electrons in covalent bonds — no free carriers except thermally gen

Fig 2.1 — Each silicon atom forms 4 covalent bonds with neighbours; the lattice is electrically neutral overall.

3. Extrinsic Semiconductor & Doping

An **extrinsic semiconductor** is one whose electrical properties have been intentionally modified by adding a controlled amount of impurity — a process called **doping**.

Purpose of doping: increase the number of free charge carriers, control conductivity, and produce either P-type or N-type material.

 **MEMORY TIP: EXTRINSIC = DOPED**

Two main types produced by doping: **P-type semiconductor** and **N-type semiconductor**.

4. P-type and N-type Semiconductors

P-type Semiconductor

- Formed by adding a **trivalent** impurity (3 valence electrons)
- Examples: Boron, Aluminium, Gallium
- One bond lacks an electron → creates a **hole**
- **Majority carriers:** Holes
- **Minority carriers:** Electrons

N-type Semiconductor

- Formed by adding a **pentavalent** impurity (5 valence electrons)
- Examples: Phosphorus, Arsenic, Antimony
- 4 electrons bond; 5th becomes a **free electron**
- **Majority carriers:** Electrons
- **Minority carriers:** Holes

A hole is *not* a physical positive particle — it represents the absence of an electron and behaves as a positive charge carrier.

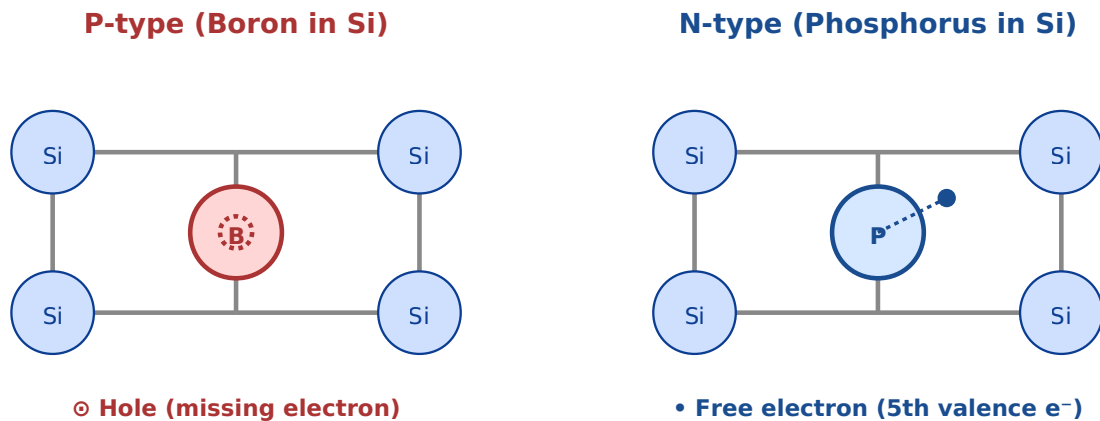


Fig 4.1 — Trivalent Boron leaves a hole in P-type silicon; pentavalent Phosphorus contributes a free electron in N-type silicon.

 **MEMORY TIP:** P-type → Positive-type → holes majority | N-type → Negative-type → electrons majority

5. Majority & Minority Carriers

Type	Majority Carriers	Minority Carriers
P-type	Holes	Electrons
N-type	Electrons	Holes

6. PN Junction Formation

A **PN junction** is formed by joining a P-type region with an N-type region. Immediately after formation, charge carriers move because of the concentration difference across the junction:

- Electrons: high concentration on N-side → move N → P
- Holes: high concentration on P-side → move P → N

This carrier movement due to concentration difference is called **diffusion**.

Diffusion

Diffusion is the movement of charge carriers from a region of **high** concentration to a region of **low** concentration, caused by the **concentration gradient**. The diffusing carriers leave behind fixed, immobile charged impurity ions near the junction.

Depletion Region

The region near the junction where mobile carriers have been depleted is the **depletion region**. It forms because:

1. Electrons diffuse $N \rightarrow P$; holes diffuse $P \rightarrow N$
2. Mobile carriers near the junction recombine
3. Fixed charged ions are left behind
4. These ions create an electric field, forming a barrier to further diffusion

Potential Barrier

The **potential barrier** is the voltage developed across the depletion region, produced by the field of the fixed ions. It opposes further diffusion of majority carriers.

Material	Approx. Barrier Potential
Silicon	$\approx 0.7 \text{ V}$
Germanium	$\approx 0.3 \text{ V}$

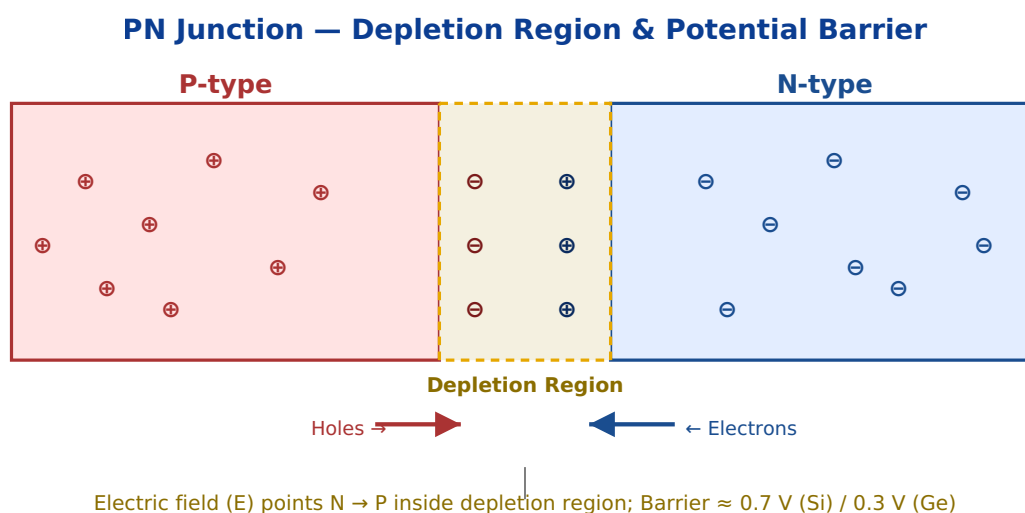


Fig 6.1 — Diffusion leaves fixed ions behind, forming the depletion region and an internal electric field that opposes further diffusion.

Drift Current

Drift current is produced by charge carrier movement under the influence of an electric field (as opposed to diffusion, which is due to a concentration gradient).

 **MEMORY TIP: DIFFUSION $\rightarrow$ CONCENTRATION GRADIENT | DRIFT $\rightarrow$ ELECTRIC FIELD**

Equilibrium

At equilibrium, diffusion tendency (driven by concentration difference) and drift tendency (driven by the junction's electric field) balance — net current is zero when no external voltage is applied.

7. Forward & Reverse Biasing

Forward Biasing

P-side connected to positive terminal, N-side to negative terminal.

- External voltage opposes the built-in barrier → barrier decreases
- Depletion region becomes thinner
- Majority carriers cross the junction more easily
- Forward current increases

Reverse Biasing

P-side connected to negative terminal, N-side to positive terminal.

- Majority carriers are pulled away from the junction
- Depletion region becomes wider; potential barrier increases
- Majority-carrier current is strongly restricted
- A very small reverse (saturation) current still flows due to minority carriers

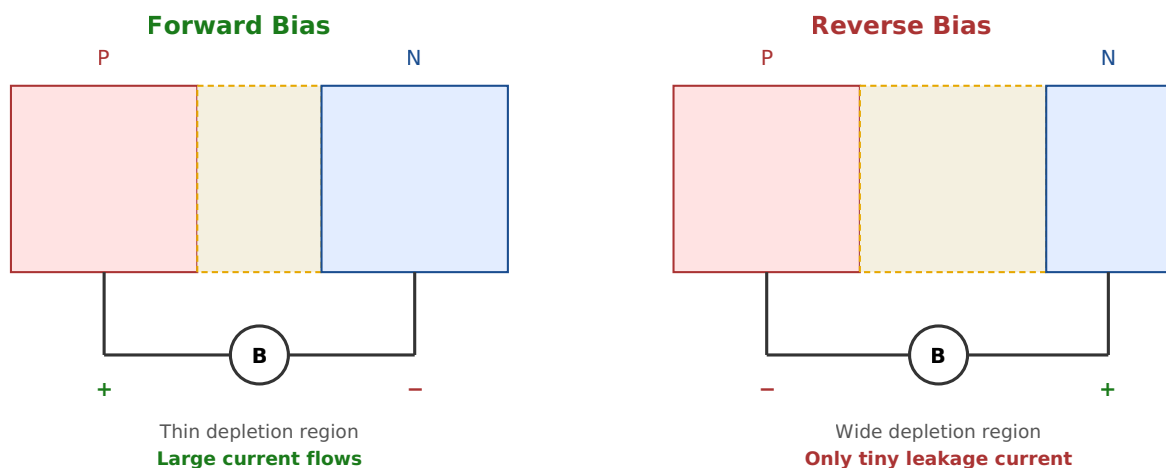


Fig 7.1 — Forward bias narrows the depletion region and allows large current; reverse bias widens it, allowing only a tiny minority-carrier current.

MEMORY TIP: FORWARD: Barrier ↓, Depletion width ↓, Current ↑ | REVERSE: Barrier ↑, Depletion width ↑, Current ≈ 0 (small)

8. V-I Characteristics of PN Junction Diode

Forward Characteristic

At small forward voltages, current is very small since the barrier still restricts carriers. As voltage approaches the **cut-in / knee voltage**, current rises noticeably; beyond the knee, a small increase in voltage causes a large increase in current.

Approximate cut-in voltage: Silicon ≈ 0.7 V, Germanium ≈ 0.3 V.

Reverse Characteristic

A very small **reverse saturation current** flows (mainly minority carriers), staying nearly constant until the **breakdown voltage** is reached, at which point reverse current increases sharply (Zener or avalanche breakdown).

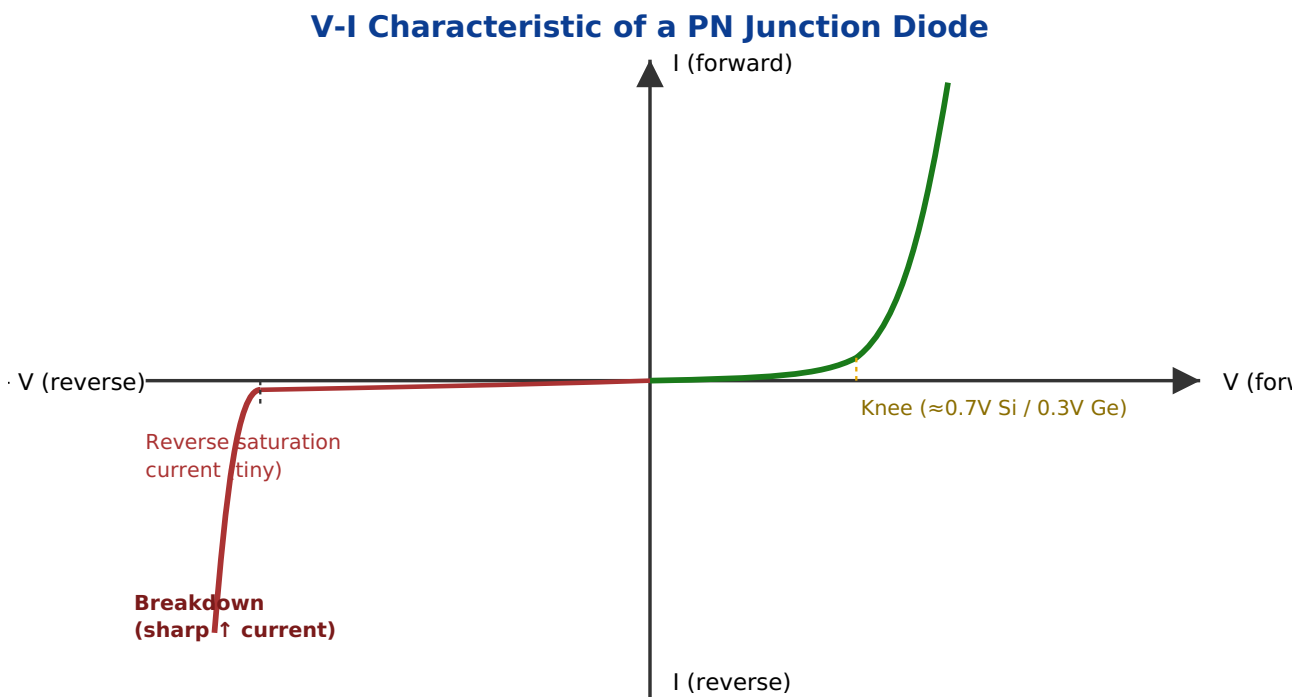


Fig 8.1 — Typical diode V-I curve: negligible current before the knee, rapid rise after; reverse side shows tiny saturation current until breakdown.

9. Breakdown Mechanisms

Breakdown occurs in reverse bias when reverse current increases sharply after the reverse voltage reaches a critical value. Two mechanisms:

Feature	Zener Breakdown	Avalanche Breakdown
Doping	Heavy	Light
Depletion region	Very thin	Wide
Mechanism	Strong field pulls carriers across junction directly	Impact ionization / carrier multiplication (collisions)
Breakdown voltage	Generally lower	Generally higher

Zener Diode

A **Zener diode** is designed to operate safely in reverse breakdown. It behaves like an ordinary diode in forward bias; in reverse bias, once the **Zener voltage (V_Z)** is reached, current rises sharply while voltage stays approximately constant — the key feature exploited for **voltage regulation**.

 **MEMORY TIP: ZENER → REVERSE BREAKDOWN → CONSTANT VOLTAGE → VOLTAGE REGULATION**

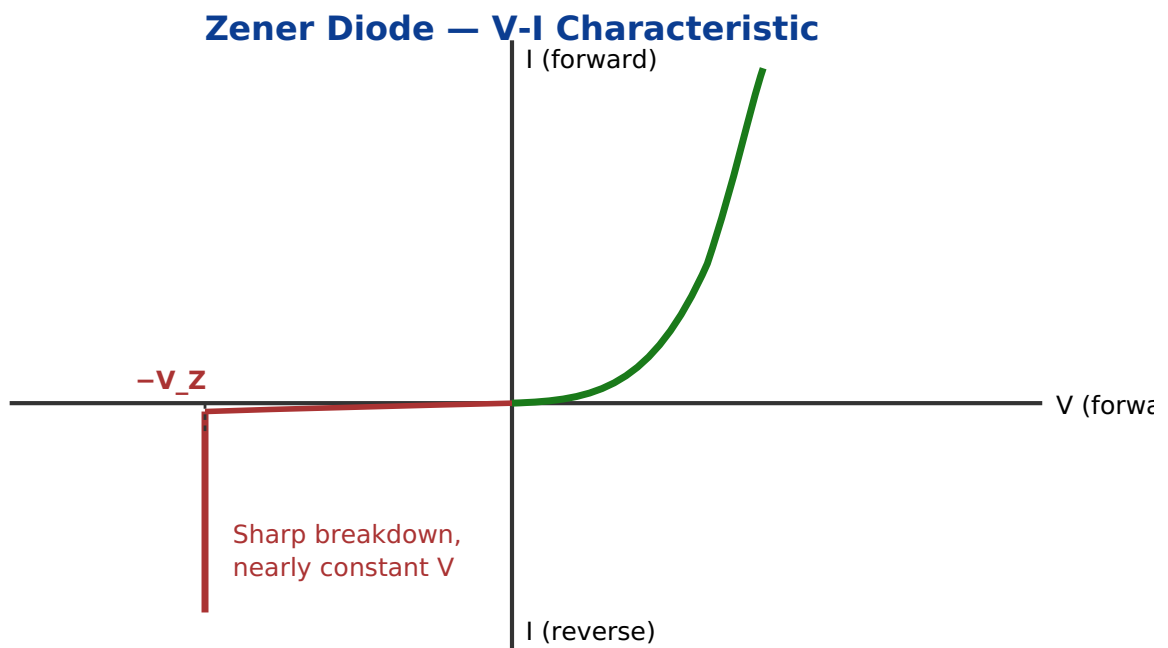


Fig 9.1 — Beyond the Zener voltage, current rises sharply while voltage across the diode stays almost constant.

10. Rectifiers

A **rectifier** converts AC into pulsating DC. A diode conducts in forward bias and blocks in reverse bias, so it selects appropriate portions of the AC waveform.

Half-Wave Rectifier

Uses **one diode**. Conducts during one half-cycle (forward biased) and blocks during the other (reverse biased) — only one half of the AC waveform appears at the output.

Half-Wave Rectifier — Circuit & Waveforms

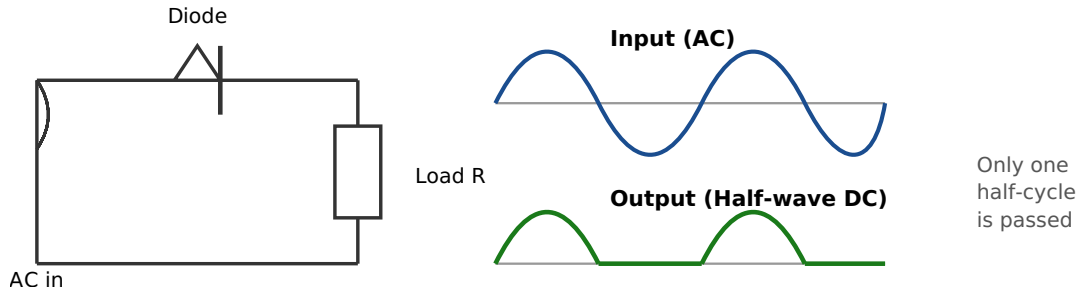


Fig 10.1 — Half-wave rectifier: only alternate half-cycles reach the load, giving pulsating DC.

Full-Wave Rectifier

Uses **both half-cycles** so current through the load flows in the same direction throughout. Two forms: **Centre-tapped** (2 diodes, centre-tapped transformer) and **Bridge** (4 diodes, no centre-tapped transformer needed).

Full-Wave Rectifier Output Waveform

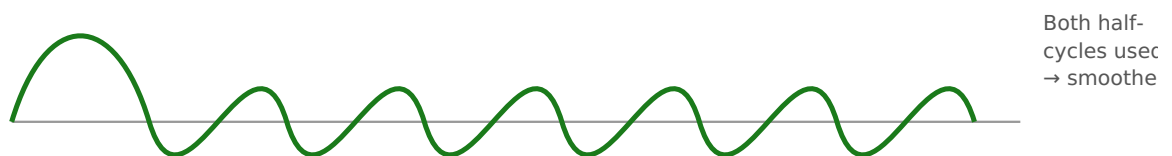


Fig 10.2 — Full-wave (centre-tapped or bridge) rectifier output: both half-cycles contribute, giving a smoother pulsating DC than half-wave.

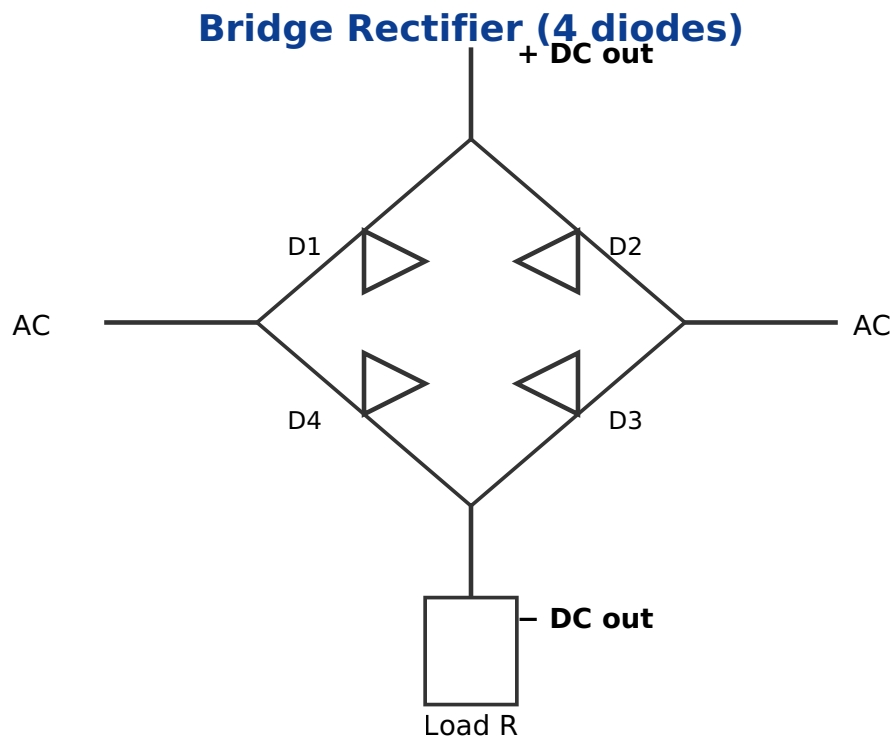


Fig 10.3 — Bridge rectifier: 4 diodes conduct in diagonal pairs each half-cycle, giving full-wave output without a centre-tapped transformer.

Rectifier Comparison

Type	Diodes	Efficiency (max)	Ripple Factor	PIV
Half-wave	1	$\approx 40.6\%$	≈ 1.21	V_m
Centre-tapped full-wave	2	$\approx 81.2\%$	≈ 0.482	$2V_m$
Bridge	4	$\approx 81.2\%$	≈ 0.482	V_m

Ripple Factor, Efficiency & PIV

- **Ripple:** unwanted AC component in the rectified output. Lower ripple factor = smoother DC.
- **Rectifier efficiency** = DC output power / AC input power (expressed as %). Full-wave is higher because it uses both half-cycles.
- **PIV (Peak Inverse Voltage):** maximum reverse voltage a diode can withstand without breakdown. The diode's PIV rating must exceed the maximum reverse voltage it will see.

11. Clipper & Clamper Circuits

Clipper — "Cuts"

Removes/clips a selected portion of a waveform (not necessarily a full half-cycle): positive, negative, or biased clipping above/below a chosen voltage.

Clamper — "Shifts"

Shifts the DC level of an AC waveform up or down using a diode, capacitor, and resistor/load, while ideally preserving its shape.

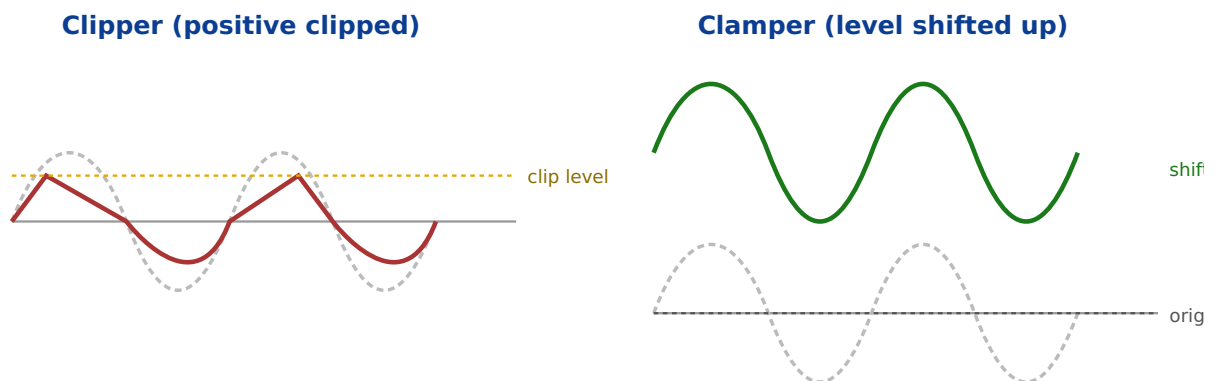


Fig 11.1 — Clipper trims part of the waveform above a set level; clamper moves the entire waveform to a new DC level while keeping its shape.

12. Voltage Doubler

Produces a DC output approximately **2×** the peak AC input, using diodes and capacitors that charge during alternate half-cycles and combine their stored voltages.

Example: if $V_m = 10\text{ V}$, ideal output $\approx 2V_m = 20\text{ V}$.

13. Special Diodes

Tunnel Diode

A **very heavily doped** PN junction with an extremely thin depletion region, allowing carriers to cross by quantum-mechanical **tunneling**. Its key exam feature is the **negative resistance region**: in this region, as voltage increases, current *decreases*. Used in high-speed switching and high-frequency oscillators.

Tunnel Diode — Negative Resistance Region

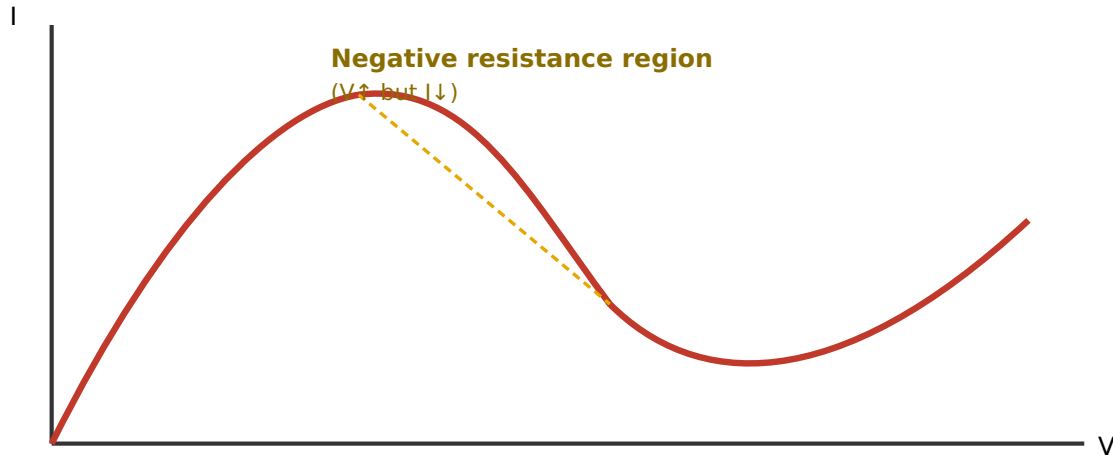


Fig 13.1 — Between the peak and valley points, current decreases as voltage increases — the negative resistance region unique to tunnel diodes.

Varactor Diode

A **voltage-controlled capacitor**, operated in **reverse bias**. As reverse voltage increases, depletion width increases, so junction capacitance **decreases** — used in tuning circuits, radio/TV tuners, and frequency control.

🧠 **MEMORY TIP: VARACTOR → VARIABLE CAPACITOR → REVERSE BIAS | Reverse V ↑ → Capacitance ↓**

14. Conceptual Chains to Memorize

A. PN Junction Formation

P region + N region → Concentration difference → Diffusion of holes & electrons → Recombination near junction → Fixed ions remain → Depletion region forms → Electric field develops → Potential barrier forms

B. Forward Bias

P → positive, N → negative → Barrier decreases → Depletion region thins → Majority carriers cross easily → Large forward current

C. Reverse Bias

P → negative, N → positive → Majority carriers pulled away → Depletion widens → Barrier increases → Only small minority-carrier current → At breakdown voltage → sharp reverse current increase

D. Zener Regulation

Reverse bias → Reach Zener voltage → Breakdown → Current changes significantly → Voltage stays ≈ constant → Voltage regulation

E. Varactor

Reverse voltage changes → Depletion width changes → Junction capacitance changes → Voltage-controlled capacitance

15. Common Mistakes to Avoid

1. P-type does **not** have electrons as majority carriers — holes are majority.
2. N-type does **not** have holes as majority carriers — electrons are majority.
3. Don't confuse diffusion (concentration gradient) with drift (electric field).
4. Potential barrier ≠ depletion region — one is a voltage, the other is a physical region.
5. Reverse bias does not mean zero current — a small reverse saturation current exists.
6. Don't confuse rectifier efficiency (AC → DC power conversion) with PIV (max reverse voltage withstood).
7. A clipper doesn't always remove a full half-cycle — it can remove a selected portion.
8. Clipper "cuts"; Clamper "shifts" — don't mix them up.
9. Zener voltage (reverse breakdown) ≠ ordinary forward cut-in voltage.
10. Avalanche breakdown involves carrier acceleration and impact ionization, not simple saturation.
11. A bridge rectifier uses 4 diodes and needs **no** centre-tapped transformer.
12. A varactor diode is normally used in **reverse** bias.
13. The tunnel diode's special feature is its **negative resistance region**.

16. Formula / Value Sheet for Exam

Quantity	Value
Half-wave max efficiency	$\approx 40.6\%$
Half-wave ripple factor	≈ 1.21
Half-wave PIV	V_m
Full-wave max efficiency	$\approx 81.2\%$
Full-wave ripple factor	≈ 0.482
Centre-tapped full-wave PIV	$2V_m$
Bridge rectifier PIV	V_m
Ideal voltage doubler output	$\approx 2V_m$
Silicon cut-in voltage	$\approx 0.7 \text{ V}$
Germanium cut-in voltage	$\approx 0.3 \text{ V}$

17. Very Short Last-Minute Revision

Term	One-line Recap
Semiconductor	Intermediate conductivity; conductivity $\uparrow$ with temperature
Intrinsic	Pure semiconductor
Extrinsic	Doped semiconductor
Doping	Controlled addition of impurities
P-type	Trivalent impurity $\rightarrow$ holes majority
N-type	Pentavalent impurity $\rightarrow$ electrons majority
PN junction	P + N joined together
Diffusion	High concentration $\rightarrow$ low concentration
Depletion region	Region depleted of mobile carriers near junction
Potential barrier	Voltage opposing majority-carrier diffusion
Forward bias	Barrier $\downarrow$, depletion $\downarrow$, current $\uparrow$
Reverse bias	Barrier $\uparrow$, depletion $\uparrow$, small reverse current
Rectifier	AC $\rightarrow$ pulsating DC
Half-wave	1 diode, $\eta \approx 40.6\%$, ripple ≈ 1.21 , $PIV = V_m$
Full-wave	Both half-cycles, $\eta \approx 81.2\%$, ripple ≈ 0.482
Bridge	4 diodes, no CT transformer, $PIV = V_m$
Ripple	Unwanted AC component in rectified output
Efficiency	DC output power / AC input power
PIV	Max reverse voltage diode can withstand
Clipper	Cuts waveform
Clamper	Shifts waveform
Voltage doubler	$\approx 2V_m$ output ideally
Zener	Reverse breakdown, voltage regulation
Avalanche	Impact ionization / carrier multiplication
Tunnel	Heavily doped, tunneling, negative resistance
Varactor	Reverse-biased voltage-controlled capacitor

— End of Notes —